

Module 3: Knowledge Representation & Logic

Propositional Logic, Entailment, CNF, First-Order Logic, Unification (MGU) & Resolution

Module III: Knowledge Representation & Logical Reasoning — Complete 9-Topic Tracker

Topic 15: Knowledge-Based Agents (KB, Inference Engine & TELL/ASK)

Topic 17: Propositional to Predicate Logic Transition

Topic 19: First-Order Predicate Logic (Constants, Predicates & Quantifiers)

Topic 21: Forward Chaining (Data-Driven Deductive Reasoning)

Topic 23: The Resolution Inference Rule (Refutation Proofs & CNF)

Topic 16: Propositional Logic (Syntax, Semantics, Truth Tables & Equivalences)

Topic 18: Propositional Logic-Based Agents (Wumpus World)

Topic 20: Knowledge Representation in FOL (English to Logic Translation)

Topic 22: Backward Chaining (Goal-Driven Hypothesis Verification)

Topic 15: Knowledge-Based Agents (KB Architecture)

A **Knowledge-Based Agent (KBA)** uses a formal, explicit internal representation of knowledge (the Knowledge Base KB) and an Inference Engine to deduce new facts and make decisions:

TELL(KB , sentence) $\leftarrow$ Adds new percept facts/rules to KB
ASK(KB , query) $\leftarrow$ Deduces whether $KB \models$ query (Entailment)

Topic 16 & 17: Propositional Logic vs. First-Order Logic (FOL)

Formal Logic System	Ontological Commitment	Epistemological Commitment
Propositional Logic	Facts that are either True or False (Boolean propositions P, Q). Treats atomic sentences as indivisible black boxes.	Belief states: {True, False, Unknown}.
First-Order Logic (FOL)	Objects (Constants), Relations (Predicates), Functions , and Quantifiers ($\forall, \exists$).	Belief states: {True, False, Unknown}. Expressive power to model complex domains.

Key Propositional Equivalence Laws: - Implication Elimination: $P \implies Q \equiv \neg P \vee Q$ - Biconditional Elimination: $P \iff Q \equiv (P \implies Q) \wedge (Q \implies P)$ - De Morgan's Laws: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$, $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Topic 19 & 20: Knowledge Representation in First-Order Logic (FOL)

Classic English → First-Order Logic Translation Suite

Natural English Statement	First-Order Predicate Logic Formulation
"Every student is intelligent."	$\forall x (\text{Student}(x) \implies \text{Intelligent}(x))$
"Some students like AI."	$\exists x (\text{Student}(x) \wedge \text{Likes}(x, \text{AI}))$
"No person can live on the sun."	$\neg \exists x (\text{Person}(x) \wedge \text{LivesOn}(x, \text{Sun})) \equiv \forall x (\text{Person}(x) \implies \neg \text{LivesOn}(x, \text{Sun}))$
"Everyone has a mother."	$\forall x \exists y \text{MotherOf}(y, x)$

Topic 21 & 22: Forward Chaining vs. Backward Chaining

Parameter	Forward Chaining (Data-Driven)	Backward Chaining (Goal-Driven)
Starting Point	Starts from known initial atomic facts in the KB.	Starts from the Goal query hypothesis Q .
Execution Direction	Applies Modus Ponens forward: $P, P \implies Q \implies \text{infer } Q$. Adds new facts until goal is reached.	Works backward: finds rules that have Q in the conclusion, and recursively proves their premises.
Suitability	Ideal for monitoring, diagnosis, and reactive synthesis systems.	Ideal for targeted automated theorem proving, diagnosis, and query answering (Prolog).

Topic 23: The Resolution Refutation Algorithm

The Resolution Inference Rule:

$$\frac{\ell_1 \vee \dots \vee \ell_i \vee \dots \vee \ell_k, \quad m_1 \vee \dots \vee \neg \ell_i \vee \dots \vee m_n}{\ell_1 \vee \dots \vee \ell_{i-1} \vee \ell_{i+1} \vee \dots \vee \ell_k \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee m_n}$$

Algorithm: Proof by Resolution Refutation ($KB \models \alpha$)

- Convert all sentences in KB into **Conjunctive Normal Form (CNF)** (Eliminate $\implies$, push $\neg$ inward, standardize variables, Skolemize $\exists$, drop $\forall$, distribute $\vee$ over $\wedge$).
- Negate the Query:** Add $\neg \alpha$ to the CNF knowledge base.
- Repeatedly apply the Resolution Rule to pairs of complementary literals (ℓ and $\neg \ell$) to generate new resolvents.
- If resolution derives the **Empty Clause** ($\square$), a formal logical contradiction is achieved $\implies$ Original query α is mathematically proven!

M3 Active Recall & Exam Questions

Q1. Given $KB = \{P \implies Q, Q \implies R, P\}$, prove R using Resolution Refutation. (8 Marks)

Step 1: Convert KB to CNF Clauses:

(1) $P \implies Q \equiv \neg P \vee Q$

(2) $Q \implies R \equiv \neg Q \vee R$

(3) P

Step 2: Add Negation of Goal R :

(4) $\neg R$

Step 3: Resolution Steps:

(5) Resolve (1) and (3) on literal $P \implies Q$

(6) Resolve (2) and (5) on literal $Q \implies R$

(7) Resolve (4) and (6) on literal $R \implies \square$ (Empty Clause / Contradiction).

$\therefore R$ is proven true by refutation!